Milestones in Autonomous Driving and Intelligent Vehicles—Part II: Perception and Planning

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***Abstract*—A growing interest in autonomous driving (AD) and intelligent vehicles (IVs) is fueled by their promise for enhanced safety, efficiency, and economic benefits. While previous surveys have captured progress in this field, a comprehensive and forward-looking summary is needed. Our work fills this gap through three distinct articles. The first part, a "survey of surveys" (SoS), outlines the history, surveys, ethics, and future directions of AD and IV technologies. The second part, "Milestones in AD and IVs Part I: Control, Computing System Design, Communication, high-definition map (HD map), Testing, and Human Behaviors" delves into the development of control, computing system, communication, HD map, testing, and human behaviors in IVs. This part, the third part, reviews perception and planning in the context of IVs. Aiming to provide a comprehensive overview of the latest advancements in AD and IVs, this work caters to both newcomers and seasoned researchers. By integrating the SoS and Part I, we offer unique insights and strive to serve as a bridge between past achievements and future possibilities in this dynamic field.**

***Index Terms*—Autonomous driving (AD), communication, control, high-definition map (HD map), human behaviors, intelligent vehicles (IVs), perception, planning, survey of surveys (SoS), system design, testing.**

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I. Introduction

AUTONOMOUS driving (AD) and intelligent vehicles (IVs) have recently attracted significant attention from academia as well as industry because of a range of potential benefits. Surveys on AD and IVs occupy an essential position in gathering research achievements, generalizing entire technology development, and forecasting future trends. However, a large majority of surveys only focus on specific tasks and lack systematic summaries and research directions in the future. As a result, they may have a negative impact on conducting research for abecedarians. Our work consists of three independent articles, including a survey of surveys (SoS) [1] and two surveys on crucial technologies of AD and IVs. Here, is the third part (Part II of the survey) to systematically review the development of perception and planning. Combining with the SoS and the second part (Part I of the survey on control, system design, communication, high-definition map (HD map), testing, and human behaviors in IVs) [2], we expect that our work can be considered as a bridge between past and future for AD and IVs.

According to the different tasks in AD, we divide them into eight sections, perception, planning, control, system design, communication, HD map, testing, and human behaviors in IVs as Fig. 1. In Part I, we briefly introduce the function of each task and the intelligent levels for AD. Here, we describe classical applications in different AD scenes, including urban roads, park logistics, warehouses, ports, intelligent mines, agriculture, wildlife rescue, and tunnel search. It is more common for citizens to realize AD in urban roads, such as private IVs, AD taxis, and buses. IVs in parts and ports require controllers to follow specific rules and achieve high efficiency. Warehouses and mines are classic closed scenes in indoor and outdoor environments. Modified IVs or called professional intelligent robots can be employed in the wild to replace the human harbor in agricultural operations, wildlife rescue, tunnel search, etc. Indeed, AD and IVs could conduct a number of tasks in different scenes and play a crucial role in our daily life.

In this article, we consider two sections as independent chapters, and each of them includes task definition, functional divisions, novel ideas, and a detailed introduction to milestones of AD and IVs, and the relationship of perception, planning, and control can be seen in Fig. 2. The most important thing is that the research of them have rapidly developed for a decade and now entered a bottleneck period. We wish this article could

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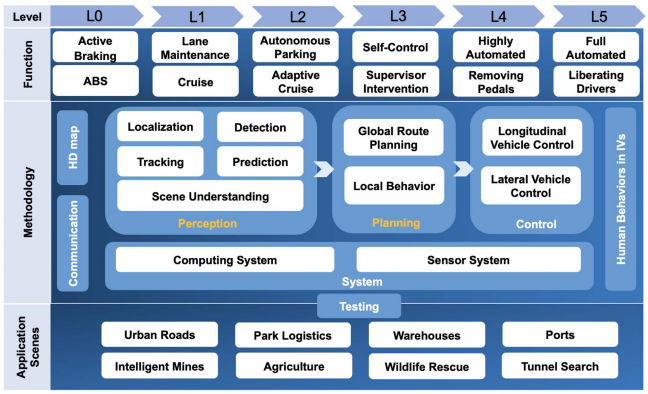


Fig. 1. Structure of AD with the function, methodology, and application scenes.

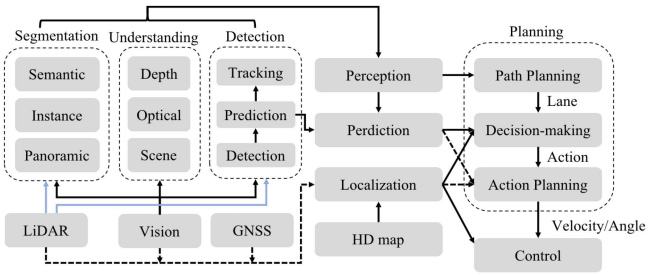


Fig. 2. Relationship of subtasks in perception, planning and the relationship of perception, and planning and control.

be considered as a comprehensive summary for abecedarians and bring novel and diverse insights for researchers to make breakthroughs. We summarize three contributions of this article.

1) We provide a more systematic, comprehensive, and novel survey of crucial technology development with milestones on AD and IVs.
2) We introduce a number of deployment details, testing methods, and unique insights throughout each technology section.
3) We conduct a systematic study that attempts to be a bridge between past and future on AD and IVs, and this article is the third part of our whole research (Part II for the survey).

II. Perception

Perception is a fundamental module for AD. This module provides surrounding environmental information to the ego-vehicle. As can be seen in Fig. 3, perception is divided into localization, object detection, scene understanding, target prediction, and tracking.

A. Localization

Localization is the technology for the driving platform to obtain its own position and attitude. It is an important prerequisite for planning and control [3]. Currently, localization strategies are divided into four categories: 1) global navigation satellite system (GNSS) and inertial measurement unit (IMU);

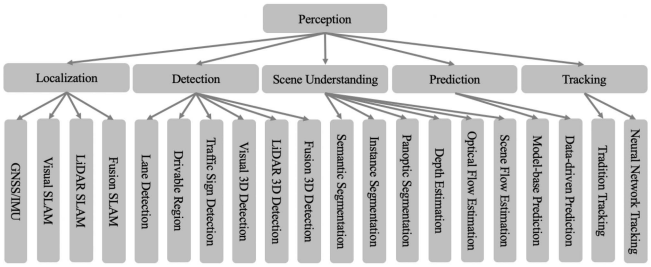


Fig. 3. Structure of perception methodology.

2) visual simultaneously localization and mapping (SLAM); 3) LiDAR SLAM; and 4) fusion-based SLAM [4].

1) GNSS and IMU: The GNSS [5] is a space-based radio navigation and localization system that can provide users with three-dimensional (3-D) coordinates, velocity, and time information on the earth's surface. The IMU [6] is commonly composed of three-axis accelerometers and gyroscopes (additional three-axis magnetometers for 9-degree-of-freedom (DOF) IMUs). By updating the localization information in low frequency from GNSS with dynamic states from IMUs, the platform could obtain the localization info in a high update frequency. Although the fusing the GNSS and IMU method is all-weather, the satellite signals will be interfered with by urban buildings [7].

2) Visual SLAM: Visual SLAM adopts the change of frames from cameras to estimate the ego-vehicle motion and this type of algorithms is divided into three categories by sensors: 1) the monocular; 2) multiply views; and 3) depth. Specifically, the visual SLAM algorithms only require images as inputs that means the cost of the localization system is relatively cheap [8]. However, they are dependent on abundant features and slight variations of illumination. In addition, optimization is a crucial module for visual localization system which updates each frame estimation after considering the global information and the optimization methods include filter-based and graph-based [9].

There are two typical categories of visual SLAM from the perspective of feature extraction, key points [10], [11], [12], [13] and the optical flow methods [14], [15], [16], [17]. The key points methods utilize points extraction approaches, like SIFT, SURF, ORB, and descriptors, to detect the same characteristics at different images and then compute relative motion among the frames. As points extraction approaches can extract crucial points stably and accurately, key points visual SLAM systems can offer significant benefits in structured roads and urban areas. However, the system may suffer difficulties when operating on an unstructured road or facing a flat white wall. Besides, earlier algorithms could not run in real-time and ignored most of the pixel information in the image. The optical flow methods assume that the photometric is invariant among the frames and attempt to estimate the camera motion by minimizing the photometric error on the images. This kind of method has several advantages as follows: 1) low-computing overhead and high-real-time performance; 2) weak dependence on key points; and 3) considering whole pixels in the frames. However, due to the photometric assumption, it is

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sensitive for the optical flow methods to the luminosity change between two images. visual SLAM systems also could be categorized into filter-based and optimization-based strategies from the optimization perspective, however, the graph-based optimization methods have made a number of breakthroughs in accuracy and efficiency. Thus, researchers will continue to focus on the latter point in the future.

3) LiDAR SLAM: Compared with the visual SLAM methods, LiDAR SLAM systems detect surrounding environments actively with accurate 3-D information because of the LiDARs properties. Similar to visual systems, the LiDAR SLAM also could be categorized into two-dimensional (2-D), such as Gmapping, Cartographer, Karto, and 3-D [18], [19], [20], [21] methods, by sensors or filter-based like Gmapping and optimization-based by the process of optimization. Gmapping adopts the particle filter approach and separates the localization and mapping processes. During the optimization, each particle is responsible for maintaining a map. LOAM [18] operates two parallel algorithms, one is to calculate the motion transformation between frames in a low frequency through point cloud matching methods, and the other attempts to construct a map and correct the odometry in a high frequency. Segmap [21] utilizes deep neural networks to extract semantic feature information, which could reduce the computational resource consumption, and solve the data compression problem in real-time for indoor intelligent robotics and IVs. SUMA [20] transfers the point clouds into 2-D space and adopts an extended RGB-D SLAM structure to generate a local map. Besides, it maintains and updates the surfel map by the iterative closest point (ICP) matching method for point clouds. LiDAR SLAM systems have the advantages of high accuracy, achieving a dense map, and weak dependency on lightness. However, no semantic information and environmental disturbance are two main challenges for LiDAR SLAM systems. In addition, researchers have to spend lots of time and effort to maintain and repair LiDARs installed on the IVs.

4) Fusion-Based SLAM: In order to avoid the problems with failures in single sensor or low robustness, fusing multiple modalities data methods have been introduced by researchers, including visual-inertial [22], [23], [24], LiDAR-inertial [25], [26], [27], [28], visual-LiDAR inertial [29], [30], [31], and other fusion, such as adding sonars [32] or radars [33], SLAM approaches. We found that the fusion methods usually introduce IMU data with higher updating frequency to SLAM systems. The loose fusion methods [18], [29] treat the external observation data from cameras or LIDARs and the internal motion data from IMUs as two independent modules, while the tight fusion approaches [24], [26], [30], [31] design a unit optimization module to solve and fuse multiple modalities data. The former methods could be considered as extended visual or LiDAR SLAM systems and are friendly for researchers to deploy on testing platforms and IVs. However, to increase the Robustness and adaptability, the tight fusion strategies provide appropriate solutions, including introducing bundle adjustment into the visual odometry system [30] and adopting association optimization [31]. In summary, the fusion-based SLAM methods solve several difficulties for a single sensor but still introduce a few challenges for jointing systems, such as calibration, synchronization, and complex processing. The advantages and disadvantages of the different methods for localization are shown in Table I.

TABLE I
ADVANTAGES AND DISADVANTAGES OF THE DIFFERENT LOCALIZATION METHODS

Method	Advantages	Disadvantages
GNSS	1.All-weather 2.Easy configuration 3.Non external sensors	1.Site requirements 2.GPS update frequency 3.High-cost (high accuracy)
Visual	1.Economy 2.Rich semantic info	1.Lightness effect 2.Similar feature 3.Low accuracy
LiDAR	1.High accuracy. 2.Dense map	1.Non semantic info 2.Environmental disturbance 3.High-cost
Fusion	1.High accuracy 2.Robustness	1.Complexity 2.Calibration 3.Synchronization

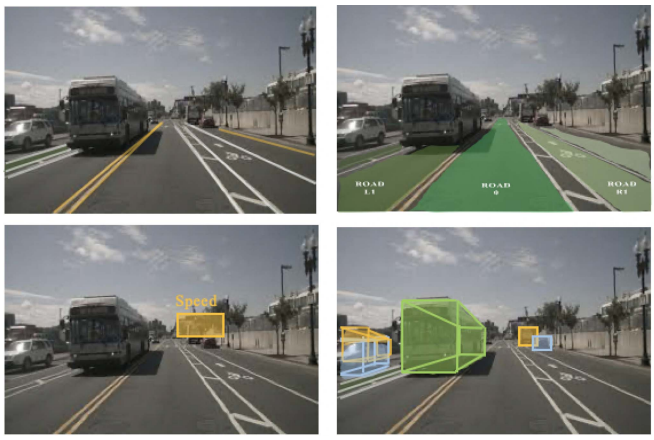


Fig. 4. Crucial tasks in the object detection for AD. The top-left, top-right, bottom-left, and bottom-right figure represents the lane detection, drivable region, traffic-sign detection, and 3-D object detection.

B. Object Detection

The purpose of object detection is to detect the static and dynamic targets in the field of view of the sensors. The results of some detection tasks can be seen in Fig. 4.

1) Lane Detection: Lane Detection is to recognize the lane in the views of sensors, to assist driving. For universal process, it involves three sections, including image preprocessing, lane detection, and tracking. The purpose of image preprocessing, such as region-of-interest (RoI) extraction, inverse perspective mapping, and segmentation, is to reduce the computing cost and eliminate noise. The methods of lane detection and tracking can be divided into the computer vision-based (CV-based) method and the learning-based method [34].

The CV-based methods in lane detection are broadly utilized nowadays, primarily because of their light computing cost and easy reproduction. A morphological top-hat transform is utilized to eliminate the irrelevant objects in the field [35]. After that, the Hough transform is applied to extract the edge pixel of the image and construct the straight lines. However, the disadvantage is that it is hard to detect the curve

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lines, so a number of researchers have introduced some effective methods on the Hough transform [35]. Some other lines estimation approaches involve the Gaussian mixture models (GMMs) [36], random sample consensus (RANSAC) [37], Kalman filter [38] in complex scenes.

The learning-based methods can be deployed on abundant scenes but they need a great deal of data to train the network with plenty of parameters. Liu et al. [39] attempted to design novel multiple subheaders structures to improve the lane detection performance. To our knowledge, lane detection is integrated into the advanced driver assistance systems (ADASs) to keep the lane or follow the former vehicle, and researchers pay more attention to 3-D lanes [40], lanes in closed areas, and unstructured roads.

2) Driving Region Detection: Driving region detection increases the obstacle information compared to lane detection, which offers the base information for obstacle avoidance function and path planning tasks. We also categorize this task into the CV-based and learning-based approaches.

Driving region detection can be converted to lane detection when the road surface is not obscured by obstacles. Otherwise, it can be seen as a combination of lane detection and 2-D target detection. When considering driving region detection as an independent task, it needs to distinguish the road pixel from targets and nondriving regions. The color histogram can meet the requirement and some researchers develop methods on color [41] and efficiency [42] to tackle the poor performance on color varying and low effect of it. The region growth methods [43] are more robust than the color histogram methods.

The learning-based methods in driving region detection are similar to image segmentation. For the machine learning algorithms, features, such as the RGB color, Walsh Hadamard, histogram of oriented gradients (HOGs), local binary pattern (LBP), Haar, and LUV channel, can be extracted by the feature extractors and the classification header, such as support vector machine (SVM), conditional random field (CRF), to obtain the final results. The deep neural network can replace the feature extractors and some improvements, such as employing the large visual regions convolutional kernels [44], connection by multiple layers [45], to achieve competitive performance. We found that learning-based driving region detection results are usually one of the branches of the scene understanding task and researchers attempt to tackle a few challenges, including 2-D–3-D transformation, complex driving regions, etc.

3) Traffic Sign Detection: Traffic signs contain plenty of crucial traffic information, such as road conditions, speed limitations, driving behavior restrictions, and other information. We also divide it into CV-based approaches and learning-based approaches.

For CV-based detection approaches, the conditions for the approximate color composition of traffic signs in a specific region (in a certain country or city) are similar. In the relatively simple original image, better results can be obtained by threshold separation of specific colors, which can be obtained by adopting the color space distribution, segmentation head, and the SVM classifier [46]. Some research develops the methods by introducing extra color channels, such as the normal RGB model, the dynamic color threshold, the probability model, and edge information. However, these approaches are hard to solve the problem caused by illumination, fading, occlusion, and bad weather. Some researchers tried to utilize the though transform on the triangle [47], circular traffic signs [48], or a coding gradient detection method [49], to handle occlusion and conjunction. The shape-based detection method can solve the problem of in-stable results caused by color change, but it has little capability to overcome the problem caused by occlusion and deformation.

The traffic sign recognition algorithm based on machine learning usually uses the sliding window method to traverse the given traffic sign image. Huang et al. [50] proposed a variant histogram feature of gradient direction, and trained a single classifier to detect traffic signs through an extreme learning machine. With the continuous research of the deep learning algorithms, more and more scholars use the deep learning algorithms to detect traffic signs. Readers can regard this classification method as handling feature extraction, including preprocessing and classification [51]. To extract deeper information, the deeper encoder, integrated space pyramid pooling (SPP) layer, cascaded RCNN network, depth separable convolution, and clipping strategy are introduced to achieve the detection accuracy and high inference speed [52]. The deep learning method has a satisfactory tolerance for the variation of the color and shape of signs, however, this type of method requires a vast amounts of data and manual annotation. Besides, detection networks should have the capability of recognizing different regions with diffident signs and detecting signs over a long distance.

4) Visual-Based 3-D Object Detection: Visual-based 3-D object detection refers to the process of acquiring 3-D information (location, dimension, and rotation) about all targets in the field from the image. We divided it into monocular-based and stereo-based detection.

a) Monocular 3-D object detection: Monocular 3-D object detection is widely developed and the accuracy has been improving in recent years. Directly associating the classification and regression methods, inheriting from 2-D object detection networks like [53], have straightforward structures but perform unsatisfied due to the ill-posed problem of recovering 3-D attributions from a single image. There are two main kinds of strategies to solve the shortcoming.

1) Some [54], [55] introduce the geometric connections between 3-D and projected 2-D candidates. For example, GS3D [56] decouples the objects into several parts to analyze the surface attributes and instance relationships. Monopair [54] and Monet3D [55] consider the relationships between the target and its two-nearest neighbors.
2) Besides regressing the 3-D bounding candidates, networks also take into account the local or full depth map [57], [58] from stereo vision or LiDAR data during the training stage.

CaDDN [57] provides a fully differentiable end-to-end approach for combining depth estimation and object detection tasks. DDMP3D [58] utilizes the feature representation of

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context and depth estimation heads to achieve competitive performance. In addition, Brazil et al. [59] introduced successive frames as inputs, which attempts to update the 3-D results by associating detection and tracking. Although these methods have no obvious advantages in accuracy, extensive academic research, and low cost make them attractive.

b) Stereo 3-D object detection: Stereo 3-D object detection approaches [60], [61], [62], [63] are inspired by the parallax analysis from binocular vision. The precise depth value can be reckoned with through the distance between the binocular centers and the associated pair of pixels. Disp-RCNN [63], OC-Stereo [61] add segmentation modules paired images from stereo cameras to induce accurate association. YOLOStereo3D [62] provides a lightweight model, outperforms a great number of stereo methods based on the complicated disparity convolution operations, and significantly reduces the length of training and testing time. In sum, the stereo-based methods could avoid the ill-posed problem of monocular images and are convenient for manufacturers to deploy and maintain in IVs, but accurate measurement on the baseline, the time cost of binocular matching, and the requirement of image preprocessing pose challenges to researchers.

5) LiDAR-Based 3-D Object Detection: LiDAR-based 3-D object detection methods recognize targets 3-D properties from point clouds data captured by LiDARs. We categorized it into voxel-wise and point-wise detection.

a) Voxel-wise object detection: The voxel-wise object detection methods represent the point cloud features in the birds eyes view (BEV) and the BEV map is divided into s series of independent voxels manually. The structural design of this type of detection network evolves from point cloud segmentation frameworks, such as PointNet [64] and PointNet++ [65], which fit the detection task at the input or output side, and its overall architecture needs to balance performance and efficiency. Taking the classic VoxelNet [66] and PointPillar [67] as examples, VoxelNet normalizes the voxels after mapping point clouds, and subsequently applies local feature extraction using several voxel feature encoding (VFE) layers to each nonempty voxel. The voxel-wise features are further abstracted by 3-D convolutional middle layers (increasing the receptive field and learning the geometric spatial representation), and finally, the object is detected and classified using a region proposal network (RPN) with position regression.

b) Point-wise object detection: Point-wise object detection, such as [68], [69], are inspired by PointNet, a classical network for indoor 6-D pose estimation with point clouds. Point-RCNN [68] is a two-stage point cloud detection framework, including candidates generation with semantic segmentation analysis at the first stage and the position revision during the second stage. VoteNet [70] extends 2-D detection structures to the 3-D framework to establish a generic detection framework for point clouds. It basically follows the PointNet++ to reduce the information loss in point cloud transformation. VoteNet also introduces a novel voting mechanism inspired by the Hough transform to locate the targets' centers instead of point on the surface, compared with other 3-D networks. It should be noticed that the number of discarded points and modality distinction due to the distance in point clouds detection should be significantly considered for researchers.

6) Fusion-Based 3-D Object Detection: LiDARs, radars, and cameras are widely deployed in IVs for perception tasks and combination of these types of sensors could make the vehicles robust and able to detect targets full-time. However, this does not mean that the fusion-based methods will outperform the approaches with a single sensor. There are two main reasons for the disadvantage of fusion-based methods: 1) it is challenging for the network to fill the modalities gap from various sensors and 2) the system error and measurement errors, such as from calibration and synchronization, are hard to eliminate and they would be propagated and amplified in the networks. Most researchers propose solutions to handle these difficulties and achieve some competitive outcomes. In this section, we categorize the fusion-based objection detection task based on the types of sensors.

a) Camera and LiDAR: Cameras and LiDARs are two crucial sensors for AD and researcher first focus on fusion parallel methods, which extract point clouds and images information at the same time. MV3D [77] and AVOD [76] utilize the shared 3-D anchors on the point cloud and the corresponding images. ContFuse [80] and MMF [81] adopt tightly coupled fusion approaches with a consecutive fusion layer. 3-D-CVF [82] introduces a cross-view spatial feature fusion method to fuse the images and point clouds. In addition, EPNet [83] focuses on the point cloud system and projects the images on it with point-based strategy on the geometric space.

Compared with parallel approaches, the sequential methods are readable and deployable because of no need to introduce association structures to reduce the gaps. F-PointNet [75] and F-ConvNet [74] attempt to reduce the searching areas by generating 3-D bounding boxes within 2-D candidates. PointPainting [84] outputs semantic information and projects each point on the corresponding point to improve 3-D object detection accuracy. CLOCs [73] fuses the data after the independent extractors and achieve a competitive result on KITTI [85]. DVF [72] adopts the 2-D truth as guidance and then extract 3-D properties by the point clouds.

b) Camera and radar: Combining the images and the data from Radars can effectively reduce the cost and maintain accuracy. Reference [86] projects radar detection results to the image space and utilize them to boost the object detection accuracy for distant objects. CRF-Net [87] develops the method with a vertical presentation.

c) LiDAR and radar: This type of fusion focuses on extremely harsh weather conditions and distinct targets. RadarNet [88] fuses radar and LiDAR data via a novel-based early fusion approach. It leverages the radar's long sensing range via an attention-based fusion. MVDNet [89] generates proposals from two sensors and then fuses region-wise features between multimodal sensor streams to improve final detection results. ST-MVDNet [90] develops the structure by enforcing output consistency between a Teacher network and a

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TABLE II
PERFORMANCE OF THE 3-D OBJECT DETECTION METHODS IN KITTI

Method	Sensors	Moderate(%)	Easy(%)	Hard(%)
VPFNet[71]	Cam+LiD	83.21	91.02	78.20
DVF[72]]	Cam+LiD	82.45	89.40	77.56
CLOCs[73]	Cam+LiD	82.28	89.16	77.23
F-ConvNet[74]	Cam+LiD	76.39	87.36	66.69
Point-RCNN[68]	LiDAR	75.64	86.96	70.70
PointPillars[67]	LiDAR	74.31	82.58	68.99
F-PointNet[75]	Cam+LiD	69.79	82.19	60.59
AVOD[76]	Cam+LiD	66.47	76.39	60.23
MV3D[77]	Cam+LiD	63.63	74.97	54.00
Disp-RCNN[63]	Stereo	43.27	67.02	36.43
YOLOStereo3D[62]	Stereo	41.25	65.68	30.42
OC-Stereo[61]	Stereo	37.60	55.15	30.25
Stereo-RCNN[60]	Stereo	30.23	47.58	23.72
MonoDETR[78]	Mono	16.26	24.52	13.93
MonoDTR[79]	Mono	15.39	21.99	12.73
CaDDN[57]	Mono	13.41	19.17	11.46
DDMP-3D[58]	Mono	12.78	19.71	9.80
GS3D[56]	Mono	2.90	4.47	2.47

Student network, and introducing missing modalities to tackle the degeneration problem when one type of data is missing.

d) Camera, LiDAR and radar: In this fusion type, researchers attempt to design a robust perception system in different weather conditions. Wang et al. [91] obtained object detection outputs with a PointNet [64] architecture by projecting the images onto the point cloud directly. Parallel to the previous frame, the point cloud from the radar is processed to predict velocity which is then associated with the final detection output. RVF-Net [92] fuses all of the data on the input procedure and achieves satisfying results on the nuScenes [93] data set.

e) Others: Ultrasonic radar judges the distance of obstacles through the time of sound transmission in the air, and the accuracy can achieve a centimeter scale within 5 m. This sensor is mostly used in autonomous parking scenes. An infrared camera with an infrared lamp can capture the infrared spectral characteristics to achieve the effect of night vision imaging. Besides, research on event cameras is one of the hot topics nowadays. Event cameras process data based on pipeline timestamps, rather than processing individual pixels in a frame plane. Because the data has the nature of time sequence, the traditional network structure cannot process the data, so how to fuse with other sensors will be one of the research points in the future.

The performance of the 3-D object detection methods with various combination of different sensor types in KITTI [85] is shown in Table II. Here, KITTI divides whole data into three evaluation scenes (easy, moderate, and hard) through the frame's complexity and computes 3-D-AP, an extended method from 2-D-AP [94] on these three scenes. We summarized as follows.

1) Adopting fusion strategies could achieve competing results for 3-D object detection tasks mainly because of introducing more initial information. But this type of method requires researchers to eliminate or reduce modal differences.
2) Due to the characteristics of sensors, limited resolutions of cameras, and the definition of the reference system in KITTI, the performance of visual-based methods is weaker than LiDAR-based. However, the visual-based methods attract a number of researchers because of their maintainability, economy, and easy deployment.
3) The self-attention mechanism (Transformer structure) and the BEV method [71], [78], [79] could improve the accuracy of cross-modality fusion, feature extraction, etc. In addition, to address data hungry and model robustness, current research studies train, and the test models on additional data, such as unScenes [93] and Waymo [95].

C. Scene Understanding

We define scene understanding in this article as the multiple outputs for each pixel or point instead of each target. This section, we divide it into three sections, segmentation, depth, and flow estimation. We only focus on academic research and applications in AD areas.

1) Segmentation in Autonomous Driving: The target of semantic segmentation is to partition a scene into several meaningful parts, usually by labeling each pixel in the image with semantics (semantic segmentation), by simultaneously detecting objects and distinguishing each pixel from each object (instance segmentation), or by combining semantic and instance segmentation (panoptic segmentation) [96]. The segmentation is one of the crucial tasks in computer vision and researchers evaluate their models on ADE20K [97], Pascal-VOC [94], CityScape [98], etc. However, in AD scenarios, the classic 3-D CV area, it is hard to complete the perception task independently. It is usually involved in lane detection, driving region detection, visual interface module, or combined with point clouds to provide semantic information. We will briefly introduce the general background based on segmentation, and then highlight segmentation research on AD.

a) Semantic segmentation: Fully convolutional network (FCN) [99] is a popular structure for semantic segmentation which adopts convolutional layers to recover the size of output maps. Some work extends FCN by introducing an improved encoder–decoder [100], the dilated convolutions [101], CRFs [102], atrous spatial pyramid pooling (ASPP) [103]. In addition, the above approaches attend to fixed, square context regions because of the pooling and dilation convolution operations. The relational context method [104] extracts the relationship between pixels. Ronneberger et al. [105] pursued high resolution by channel concatenation and skip connection, especially in the medical field. In the field of AD, the semantic segmentation networks may be familiar with the common structures, and researchers should pay more attention to the special categories, and occlusion, and evaluate their models on data sets of road scenarios [98]. To achieve SOTA results on data sets, the researcher introduces the multiple scale attention mechanism [106], boundary-aware segmentation module [107]. Besides, some research focuses on the targets' attributes on roads like considering the intrinsic relevance among the cross-class objects [108] or semi-supervised segmentation mechanism because of the lack of labeled data on AD scenarios.

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b) Instance segmentation: Instance segmentation is to predict a mask and its corresponding category for each object instance. Early method [115] designs an architecture to realize both object detection and segmentation missions. Mask-RCNN [115] extends Faster-RCNN to identify each pixel's category with binary segmentation and pools image features from RoI following an RPN. Some researchers develop the base structure by introducing a coefficients network [116], the IoU score for each mask, and shape priors to refine predictions. Similar with the 2-D object detection methods, Chen et al. [117] replaced the detectors with the one-stage structures. Wang et al. [118] attempted to avoid the effect of detection and achieve remarkable performances. To achieve competitive segmentation results on AD datasets, researchers focus on the geometric information on 3-D space [119], boundary recognition [120], combining the semantic segmentation (panoptic segmentation) [121], or intruding multiple frames (video-base) [122].

c) Panoptic segmentation: Panoptic segmentation is proposed to unify pixel-level and instance-level semantic segmentation [123], and [124] designs a different branch to regress the semantic and instance segmentation results. Panoptic-FCN [125] aims to represent and predict foreground things and background stuff in a unified fully convolutional pipeline. Panoptic SegFormer [126] introduces a concise and effective framework for panoptic segmentation with transformers. For AD scenarios, TASC [110] proposes a new differentiable approach to reduce the gap between the two subtasks during training. Axial-DeepLab [109] builds a stand-alone attention model with a global receptive field and a position-sensitive attention layer to capture the positional information with low-computational cost. Besides, researchers address the multiple scales on roads by introducing a novel crop-aware bounding box regression loss and a sample approach [127], and capture the targets' boundary by a combinatorial optimization strategy. These methods achieve competitive results on the task of CityScape [98] or Mapillary Vistas [128].

2) Depth Estimation in Autonomous Driving: This type of task is to present the depth information on the camera plane, which is an effective way to enhance the visual-based 3-D object detection and a potential bridge to connect the LiDAR and camera.

The depth completion task is a subproblem of depth estimation [129]. In the sparse-to-dense depth completion problem, researchers infer the dense depth map of a 3-D scene from a sparse depth map by computational methods or multiple data from sensors. The main difficulties include: 1) the irregularly spaced pattern in the sparse depth; 2) the fusion methods for multiple sensor modalities (optional); and 3) the lack of dense pixel-level ground truth for some data and the real world (optional).

Depth estimation is the task of measuring the distance of each pixel relative to the camera. The depth value is extracted from either monocular or stereo images with supervised (the dense map obtained by depth completion) [130], unsupervised [131], LiDAR guidance [132], or stereo computing [133]. Some approaches [134], [135] introduce the CRF module, multitasks structure, global extractor, and the piece-wise planarity priors to achieve competitive performances in popular benchmarks, such as KITTI [85] and NYUv2 [136]. Models are typically evaluated according to an RMS metric [85].

For outdoor monocular depth estimation, DORN [137] adopts a multiscale network structure to capture the contextual information. MultiDepth [114] makes use of depth interval classification as an auxiliary task. HGR [138] proposes a hierarchical guidance and regularization learning framework to estimate the depth. SDNet [113] improves the results by utilizing a dual independent estimation head involving depth and semantics. VNL [112] designs a novel structure that includes local planar guidance layers at multiple stages. Yin et al. [139] used the geometric constraints of normal directions determined by randomly sampled three points to improve the depth prediction accuracy. BANet [111] introduces bidirectional attention modules which adopt the feed-forward feature maps and incorporate the global information to eliminate ambiguity. The Unsupervised method [140] attracts plenty of researchers because it could reduce the requirements on the labeled data and eliminate the over-fit problem. In addition, the pure monocular depth estimation only obtains the relative depth value because of the ill-posed problem, and the stereo guidance methods could obtain the absolute depth value. Li et al. [141] introduced the Transformer structures to achieve competitive results. The stereo depth estimation methods can be found in the stereo disparity estimation task.

3) Flow Estimation in Autonomous Driving: Similar to the segmentation and depth estimation tasks, flow estimation focuses on the image plane and it presents the pixel movement during a data frame. It attracts interest nowadays and its research can be used in event camera methods.

a) Optical flow estimation: Optical flow refers to the pixels' movement in the imaging system, including two directions, the horizontal, and vertical. Similar to unsupervised video-based depth estimation, the pixel motion [142] can be deduced by minimizing differences between the target and source images. SPyNet [143] proposes a lightweight framework that adopts classical spatial-pyramid formulation for optical flow estimation. In addition, it attempts to estimate large-displacement movement and accurate subpixel flow. PWC-Net [144] includes three subnets, the feature pyramid extractor, warping layer, and cost volume layer, to improve the quality of optical flow.

b) Scene flow estimation: Scene flow estimation indicates a 3-D movement field which can be treated as the extension of optical flow. Therefore, it is the combination of optical flow and depth estimation in 3-D scenarios. Monocular images are seldom utilized in the holistic training step for scene flow, and the structure takes the binocular videos as input to regress disparity to restore the scale. DRISF [145] treats the inference step of Gaussian newton (GN) as a recurrent neural network (RNN) which means it can be trained in an end-to-end method. FD-Net [146] further extends the unsupervised depth estimation and disentangles the full flow into object flow (targets pixels) and rigid flow (background pixels) to assess the characteristics, respectively, which is able to avoid the warping

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TABLE III
RESULTS ON CITYSCAPE FOR PANOPTIC SEGMENTATION AND KITTI FOR DEPTH ESTIMATION

Method	PQ(%)	SQ(%)	RQ(%)
Axial-D[109]]	62.7	82.2	75.3
TASC[110]	60.7	81.0	73.8
Method	SILog(log(m))	sqErrorRel(%)	iRMSE(1/km)
BANet[111]	0.1155	2.31	12.17
VNL[112]	0.1265	2.46	13.02
SDNet[113]]	0.1468	3.90	15.96
MultiDepth[114]	0.1605	3.89	18.21

ambiguity due to the occlusion and truncation. Competitive collaboration (CC) [147] sets the scene flow estimation as a game with three players. Two of them compete for the resource and the last one acts as a moderator. GeoNet [148] consists of two modules, a monocular depth with the six DoF ego-motion estimation, and a residual network to learn the object's optical flow.

The performance of panoptic segmentation and depth estimation on CityScape and KITTI is shown in Table III. PQ, SQ, RQ refer the panoptic segmentation, segmentation quality, and recognition quality, respectively, in [123], and for depth estimation, Scale invariant logarithmic error (SILog), Relative squared error (sqErrorRel), and Root mean squared error of the inverse depth (iRMSE) are classical metrics in KITTI. Similar to detection, researchers introduce the self-attention mechanism, extra training data and novel network units to develop the accuracy in scene understanding tasks. And we mention that above tasks do not directly provide outputs to the downstream tasks such as planning and control in AD. In the actual tasks, semantic segmentation, depth estimation, and optical flow estimation will be combined with each other to provide richer pixel semantic information, so as to improve the accuracy of cross-modality data fusion, spatial detection, and tracking for moving targets.

D. Prediction

In order to safely and efficiently navigate in complex traffic scenarios, an AD framework should be able to predict the way in which the other traffic agents (such as vehicles and pedestrians) will behave in the near future. Prediction can be defined as probable results according to past perceptions. Let X_t^i be a vector with the spatial coordinates of agents i at observation time t, with $t \in \{X_1^i, X_2^i, \ldots, X_{T_{obs}}^i\}$.

1) Model-Based Approaches: These methods predict the behaviors of agents, such as changing lanes, turning left, and so on. One of the simplest methods to predict the probability distribution of vehicle behavior is the autonomous multiple models (AMMs) algorithm. This algorithm computes the maximum probability trajectory of each agent.

2) Data-Driven Approaches: These methods are mainly composed of the neural network. After training on the perception dataset, the model makes a prediction of the next behavior. DESIRE [149] proposes an encoder–decoder framework that innovatively incorporates the scenario context and the interactions between traffic agents. SIMP [150] discretizes the output space, calculates the distribution of the vehicle's destination, and predicts an estimated time of arrival and a spatial offset. FaF [151] pioneers the unification of detection and short-term motion forecasting based on LiDAR point clouds. The prediction module is sometimes separated from the perception, mainly because the downstream planning module receives both the perception and the prediction results. Future research on prediction will focus on the formulation of generalized rules, the universality of scenarios, and the simplicity of modules.

E. Tracking

The tracking problem begins with a sequence of vehicle-mounted sensor data. Depending on if the neural network is embedded in the tracking framework, we divide them into the traditional method and the neural network method.

1) Tradition Method: The Kalman filter [152] is a famous algorithm, particularly with regard to tracking agents. Because of the low-computational cost, the Kalman-based method [153] has a quick response time even on low-spec hardware in simple scenarios.

The tracking problem also can be shown as a graph search problem [154]. Compared with Kalman-based methods, The most important advantage of the graph-based approach is that it is better for multitracking problems. Yan et al. [155] exploited the graph-based methods using the min-cost approach to solve tracking problems.

2) Neural Network Method: Neural networks have the advantage of being able to learn important and robust features given training data that is relevant and with sufficient quantity.

CNN is widely used in agents tracking. Zhu et al. [156] handled multiagent tracking using combinations of values from convolutional layers. Hoffmann et al. [157] proposed appropriate filters for information drawn from shallow convolutional layers, achieving the same level of robustness compared with deeper layers or a combination of multiple layers.

RNN also provides a smart method to solve temporal coherence problem in tracking task. Milan et al. [158] used an LSTM-based classifier to track agents across multiple frames in time. Compared with the CNN method, the LSTM-based approach is better suited to remove and reinsert candidate observations particularly when objects leave or reenter the visible area of the scene. Joint perception and tracking can achieve the SOTA results in these two tasks. In reality, stable tracking can reduce the requirements of the system for real-time detection and can also correct the detection results. At present, the strategy of joint task learning has been favored by more and more researchers.

III. PLANNING

The planning module is responsible for finding a local trajectory for the low-level controller of the ego vehicle to track.

The planning module is responsible for finding a local trajectory for the low-level controller of the ego vehicle to track. Herein, "local" means that the resultant trajectory is short in its spatial or temporal range; otherwise, the ego vehicle cannot react to risks beyond the sensor ranges. The planning

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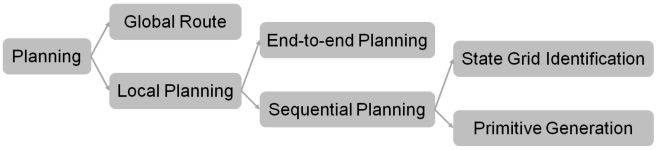


Fig. 5. Structure of the planning methodology.

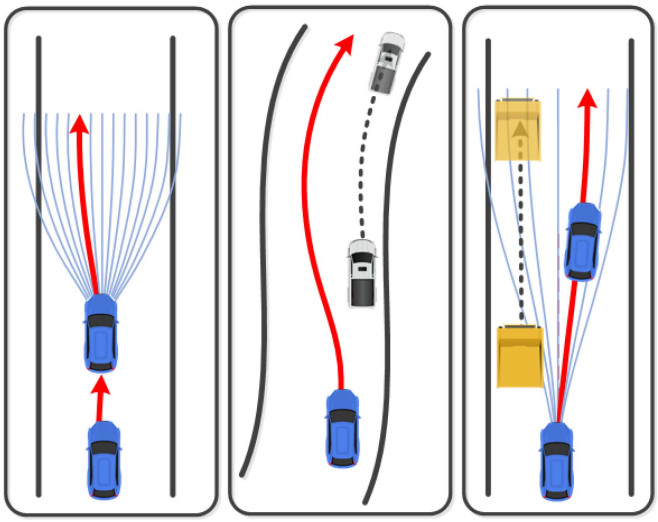


Fig. 6. Route section for local behavior planning.

module typically contains three functions, namely, global route planning, local behavior planning, and local trajectory planning [159]. Global route planning provides a road-level path from the start point to the destination on a global map; local behavior planning decides a driving action type (e.g., car-following, nudge, side pass, yield, and overtake) for the next several seconds while local trajectory planning generates a short-term trajectory based on the decided behavior type. This section reviews the techniques related to the three functions in the planning module as Fig. 5.

A. Global Route Planning

Global route planning is responsible for finding the best road-level path in a road network, which is presented as a directed graph containing millions of edges and nodes. A route planner searches in the directed graph to find the minimal-cost sequence that links the starting and destination nodes. Herein, the cost is defined based on the query time, preprocessing complexity, memory occupancy, and/or solution robustness considered.

The development history of global route planning techniques is much longer than that of autonomous vehicle techniques because global route planning also serves manually driven vehicles. As indicated by [160], the existing global route planning methods are classified as goal-directed methods, separator-based methods, hierarchical methods, bounded-hop methods, and their combinations.

B. Local Behavior/Trajectory Planning

Local behavior planning and local trajectory planning functions work together to output a local trajectory along the identified global route as shown in Fig. 6. Since the resultant trajectory is local, the two functions have to be implemented in a receding-horizon way unless the global destination is not far away [161]. It deserves to emphasize that the output of the two functions should be a trajectory rather than a path [162], otherwise, extra efforts are needed for the ego vehicle to evade the moving obstacles in the environment.

Broadly speaking, the two functions would work in two different ways. One is the end-to-end way, i.e., to develop an integrated system that receives the raw data from the on-board sensors and outputs a local trajectory directly. The other way is to implement the local behavior planning and local trajectory planning functions sequentially.

1) End-to-End Methods: Compared with the sequential-planning solution reviewed in the next section, an end-to-end solution nominally deals with vehicle-environment interactions more efficiently because there is not an external gap between the perception and planning modules [163]. The input of an end-to-end system is the large amount of raw data obtained by the on-board sensors whereas the output is a local trajectory. Since the relationship between the input and output is too intricate to be summarized as complete rules [164], the machine learning methods are commonly used, most of which are classified as the imitation-learning-based and reinforcement-learning-based methods [165].

An imitation-learning-based method builds a neuro network based on training samples [166], [167]. Challenges lie in how to collect massive training samples that are consistent and how to guarantee learning efficiency (e.g., free from overfitting). The reinforcement-learning-based methods obtain knowledge by trial-and-error operations, thus they rely less on the quality and quantity of external training samples [168].

The end-to-end methods are still not mature, thus most of them are trained/tested in simulations rather than real-world scenarios. Recent research efforts focus on how to enhance learning interpretability, security, and efficiency.

2) Sequential-Planning-Based Methods: As opposed to the aforementioned end-to-end solution, applying local behavior planning, and trajectory planning functions sequentially has been a common and conventional choice in the past decade. However, the boundary between local behavior planning and trajectory planning is rather blurred [159], e.g., some behavior planners do more than just identify the behavior type. For the convenience of understanding, this article does not distinguish between the two functions strictly, and the related methods are simply regarded as the trajectory planning methods.

Nominally, trajectory planning is done by solving an optimal control problem (OCP), which minimizes a predefined cost function with multiple types of hard constraints satisfied [169]. The solution to the OCP is presented as time-continuous control and state profiles, wherein the desired trajectory is reflected by (part of) the state profiles. Since the analytical solution to such an OCP is generally not available, two types of operations are needed to construct a trajectory.

Concretely, the first type of operation is to identify a sequence of state grids while the second type is to generate primitives between adjacent state grids.

a) State grid identification: State grid identification can be done by search, selection, optimization, or potential

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minimization. The search-based methods abstract the continuous state space related to the aforementioned OCP into a graph and find a link of states there. The prevalent search-based methods include A* search [170] and dynamic programming (DP) [171]. The selection-based methods decide the state grids in the next one or several steps by seeking the candidate with the optimal cost/reward function value. The Greedy selection [172] and the Markov decision process (MDP) series methods typically [173] fall into this category. An optimization-based method discretizes the original OCP into a mathematical program (MP), the solution of which are high-resolution state grids [174], [175]. MP solvers are further classified as gradient-based and nongradient-based ones; gradient-based solvers typically solve nonlinear programs [169], [175], quadratic programs [176], [177], quadratically constrained quadratic programs [178] or mix-integer programs [179]; nongradient-based solvers are typically represented by metaheuristics [180]. Potential-minimization-based methods adjust the state grid positions by simulating the process they are repelled or attracted by forces or in a heuristic potential field. Prevalent methods in this category include the elastic band (EB) series [181], artificial potential field methods [162], and force-balance model [182].

The capability of each state grid identification method is different. For example, gradient-optimization-based and potential-minimization-based methods are generally more flexible and stable than typical search-/selection-based methods [183], but search-/selection-based methods are more efficient to explore the entire state space globally [181], [184], [185]. Different methods could be combined jointly as a coarse-to-fine strategy, as has been implemented by many studies such as [169], [175], [176], [178].

b) Primitive generation: Primitive generation is commonly done by closed-form rules, simulation, interpolation, and optimization. Closed-form rules refer to methods that generate primitives by the analytical methods with closed-form solutions. The typical methods include the Dubins/Reeds-Shepp curves [186], polynomials [172], and theoretical optimal control methods [187]. The simulation-based methods generate trajectory/path primitives by forward simulation, which runs fast because it has no DOF [188]. The interpolation-based methods are represented by splines or parameterized polynomials. The optimization-based methods solve a small-scale OCP numerically to connect two-state grids [189].

State grid identification and primitive generation are two necessary operations to construct a trajectory. Both operations may be organized in various ways. For example, Kuwata et al. [188] integrated both operations in an iterative loop; Rufli and Siegwart [189] built a graph of primitives offline before online state grid identification; Fan et al. [176] identified the state grids before generating connective primitives.

If a planner only finds a path rather than a trajectory, then a time course should be attached to the planned path as a post-processing step [190]. This strategy, denoted as path velocity decomposition (PVD), has been commonly used because it converts a 3-D problem into two 2-D ones, which largely facilitates the solution process. Conversely, the non-PVD methods directly plan trajectories, which has the underlying merit to improve the solution optimality [171], [191], [192].

Recent studies in this research domain include how to develop specific planners that fit specific scenarios/tasks particularly [161], and how to plan safe trajectories with imperfect upstream/downstream modules [193].

IV. Conclusion

This article is the third part of our work (Part II for the technology survey). In this article, we provide a review of wide introductions on research development with milestones of perception and planning in AD and IVs. In addition, we provide a few experiment results and unique opinions for these two tasks. In combination with the other two parts, we expect that our whole work will bring novel and diverse insights to researchers and abecedarians, and serve as a bridge between past and future.

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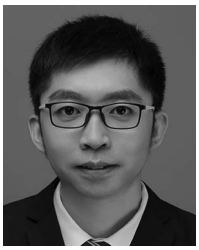
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